

Consider a partition of unity (Pou)

$$\lambda_i(x) : \mathbb{R}^d \rightarrow \mathbb{R}$$

$$\text{for all } x \quad \lambda_i \geq 0$$

$$\sum_i \lambda_i = 1$$

Then Pous are closed under convex combination
(aka a convex comb of Pous is a Pou)

Pf Let $W \in \mathbb{R}^{n \times m}$

$$W_{ij} \geq 0$$

$$\sum_j W_{ij} = 1$$

$$\text{and } \psi_i^0(x) = \sum_j W_{ij} \lambda_j(x)$$

Pou?

$$\text{Clearly } \psi_i^0 \geq 0$$

$$\begin{aligned} \sum_i \psi_i^0 &= \sum_{i,j} W_{ij} \lambda_j \\ &= \sum_j \left(\sum_i W_{ij} \right) \lambda_j \\ &= \sum_j \lambda_j = 1 \end{aligned}$$

Recall $\psi'_{ij} = \psi_i^0 \frac{d}{dx} \psi_j^0 - \psi_j^0 \frac{d}{dx} \psi_i^0$

$$\frac{d}{dx} W^0 \subseteq W'$$

Search for

$$F = -\nabla u + N[u]$$

$$\nabla \cdot F = f$$

or in weak form

$$\begin{aligned} (F, v) + (\nabla u, v) - (N[u], v) &= 0 \\ - (F, \nabla z) &= (f, z) \end{aligned}$$

$$\forall v \in W', z \in W^0$$

Denote this system $\boxed{R(u, F) = 0}$

First interaction with graph calculus

def k -clique nodes, ordered anti-symmetric k -tuple of WRT swap

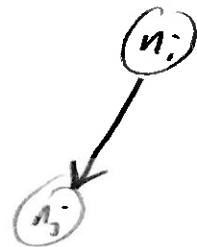
$$w \in C_k \rightarrow w = (n_i, n_j, \dots)$$

$$C_0 \xleftarrow{\partial} C_1$$

ex

$$e_{ij} = (n_i, n_j) \in C_1$$

$$(n_j, n_i) = -(n_i, n_j) \in C_1$$



$$\partial(n_i, n_j) = n_j - n_i$$

def
def

graph functions $f_i, g_i = -g_{ji}$

Graph gradient acting on graph-valued functions

$$\delta_0 f_{ij} = f_j - f_i$$

Can write as matrix

$$(\delta_0)_{e_{ij}, a, b} = \delta_{jb} - \delta_{ia}$$

$$N \begin{array}{c} \xrightarrow{\delta_0} \\ \xleftarrow{\delta_0^*} \end{array} E$$

def graph divergence as adjoint

$$\forall f, g \quad (f, \delta_0^* g) = (\delta_0 f, g)$$

$$(\delta_0 f, g) = \frac{1}{2} \sum_{i,j} (f_j - f_i) g_{ij}$$

$$= \frac{1}{2} \sum_{i,j} f_j g_{ij} - \frac{1}{2} \sum_{i,j} f_i g_{ij}$$

$$= \frac{1}{2} \sum_{i,j} f_i g_{ji} - \quad "$$

$$= -\frac{1}{2} \sum_{i,j} f_i g_{ij} - \quad "$$

$$= \sum_i \left(\sum_j -g_{ij} \right) f_i$$

$$= (\delta_0^* g, f)$$

Define the graph gradient

$$\delta_0 f_{ij} = f_j - f_i$$

Etymology

$$\int_{e_{ij}} \nabla \phi \cdot dl = \phi_j - \phi_i$$

Note

$$\delta_0^T u = \sum_{j \sim i} u_{ij}$$

Can write as matrix

$$(\delta_0)_{e_{ij}, ab} = \delta_{jb} - \delta_{ia}$$

In bilinear form we need to compute

$$(u', v) = \underbrace{\hat{v}^T (\psi^0, \psi^0)}_{D_{ij}} \hat{u}$$

— Define M_0, M_1

Thm

$$D_{ij} = M_1 \delta_0$$

pf

$$\frac{d}{dx} u = \sum_i \hat{u}_i \frac{d}{dx} \psi_i^0$$

$$= \sum_i \hat{u}_i \left(\sum_j \psi_j^0 \frac{d}{dx} \psi_i^0 - \psi_i^0 \frac{d}{dx} \psi_j^0 \right)$$

$$= \sum_i \left(\sum_j \hat{u}_j \right) \psi_{ij}' \quad \psi_{ij}'$$

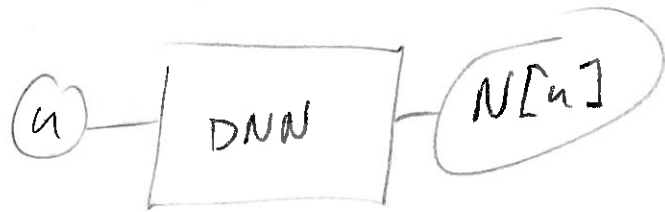
$$(v, \frac{d}{dx} u) = \hat{v}^T (\psi', \sum_j \psi_{ij}') \hat{u}_i$$

$$= \hat{v}^T (M_1 \delta_0 \hat{u})$$

How to pose trainable nonlinearity?

$(N[u], v)$, if $N[u]$ were defined

as



we would not be able to integrate it exactly, breaking the theory

Instead $\rightarrow N[u] = \sum_{i,j} \widehat{N[u]}_{i,j} \psi_{i,j}^1$

$$\rightarrow (N[u], v) = \hat{v}^T M, \widehat{N[u]}$$

Giving us

$$\begin{pmatrix} M_1 & M_1 \delta \\ -\delta^T M_1 & 0 \end{pmatrix} \begin{pmatrix} \hat{F} \\ \hat{u} \end{pmatrix} + \begin{pmatrix} M_1 N(\hat{u}) \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Back substituting

$$-\delta^T M_1 \delta \hat{u} + \delta^T M_1 N(\hat{u}) = \text{RHS}$$

Remark

Discrete

divergence

$\Rightarrow$

conservation

Note

Extracting

mass

matrices

$$\text{Let } \hat{M}_0 = (\phi_i, \phi_j), \quad \psi_i^0 = \sum_j W_{ij} \phi_j$$

$$\begin{aligned} \text{Then } M_0 &= (W\phi, W\phi) \\ &= W^T (\phi, \phi) W \\ &= W^T \hat{M}_0 W \end{aligned}$$

We can get M_1 through a similar formula

$$M_1 = (W \otimes W)^T \hat{M}_1 (W \otimes W)$$

Can we solve

$$\min \|u - u_{\text{data}}\|^2$$

$$\text{s.t. } R(u, F) = 0$$

only if R is solvable

Recall for

$$Ax + Fx = b$$

We needed $\|A^{-1}\| \leq C$

→ this is what we shared last lecture